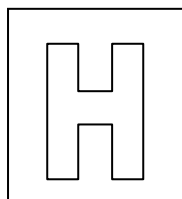


Candidate Name: _____

Class Adm No

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2021 End-of-Year Exams Pre-University 2

BIOLOGY

8876/02

Paper 2 Structured and Free-response Questions

15 September 2021

2 hours

Candidates answer on the Question Paper.
No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 26
2	/ 10
3	/ 9
Section B	/ 15
Total	/ 60

This question paper consists of 24 printed page, including 3 blank pages.

[Turn over

Section A [Total: 45 marks]

Answer **all** questions in this section.

- 1.** Proteins are one of the biomolecules of life that play various roles in a cell.

(a) State two other biomolecules that can be found in cells.

.....[1]

(b) Enzymes are a common example of proteins found in cells. Enzymes are known to denature at temperatures beyond their optimum temperature.

(i) Explain how high temperature causes enzymes to denature.

.....
.....
.....
.....
.....
.....[3]

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At high temperatures beyond optimum temperatures, enzymes show decreased to no activity due to denaturation.

At 0°C, most enzymes tends to show relatively low to no activity at all. However, it is not considered denatured, but rather **inactive**.

At sub-zero temperatures (below 0°C), enzyme activity is completely stopped due to '**cold denaturation**'.

A group of researchers used computer simulation to investigate how protein molecules like enzymes behaved in an aqueous environment at three different temperatures:

- sub-zero
- optimum
- beyond optimum

Fig. 1.1 shows the results.

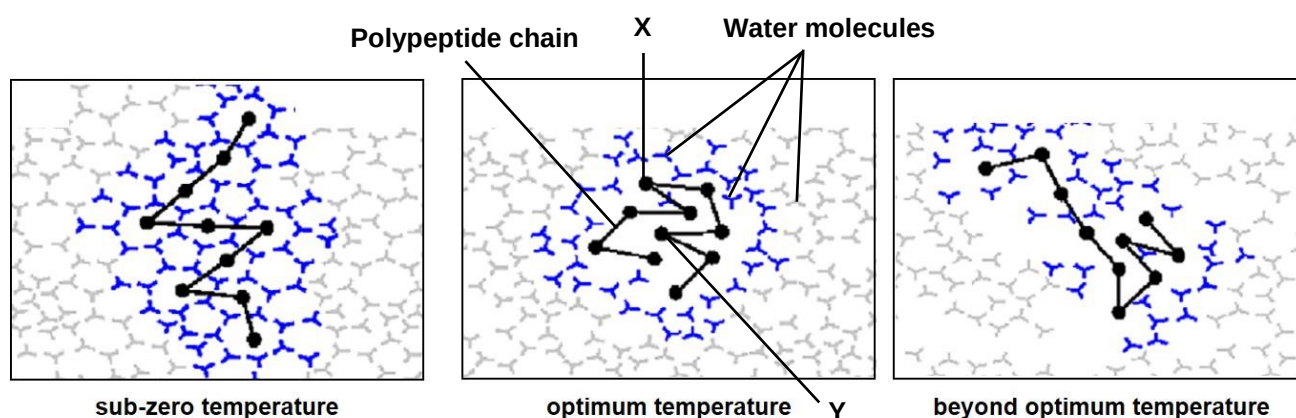


Fig. 1.1

X and Y represents amino acids in a polypeptide chain.

(ii) State the type of R-groups amino acids X and Y would likely to have.

X:

Y:

[2]

(iii) With reference to the information given and Fig. 1.1, compare the **structure** and **activity** of the enzyme when it is considered:

- 'inactive' and
- after undergoing 'cold denaturation'

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.....[3]

Freezing temperatures can also damage the cell surface membrane and membranes within the cell.

(c) (i) State an example of membranes found within the cell.

.....[1]

(ii) With reference to the structure and properties of membranes, explain why the transport of large molecules would no longer be possible at freezing temperatures.

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.....[2]

A method called *in vitro* translation is often used by scientists to produce proteins in laboratory. The proteins produced can then be purified and isolated for use in subsequent experiments. This method uses extracts from animal cells, plant cells or bacteria. The method involves the following steps:

- select for cells that have high levels of protein synthesis.
- cells are treated so that cell walls, if present and cell membranes are broken down.
- cell extract is then treated to destroy cell's own DNA and mRNA.
- desired mRNA from any source is then added to these extracts, it will be translated into the corresponding protein.

(d) Suggest a reason why:

- (i)** the cell's own mRNA needs to be destroyed.

.....
[1]

- (ii)** mRNA from any source can be translated in any type of extract.

.....

[2]

(e) Describe two differences between the cell structures used in translation in prokaryotes and eukaryotes.

.....

[2]

Proteins play a variety of key roles in cellular functions. Defective proteins due to mutations can affect various cellular processes and lead to diseases like cancer.

Mutations can be classified as gene mutation or chromosome aberration.

(f) (i) Fill in Table 1.1 to identify:

- One mechanism/cause of gene mutation and chromosome aberration
- The **corresponding effect** of the mechanism/cause of the gene mutation and chromosome aberration

Table 1.1

	mechanism/cause	corresponding effect
gene mutation		
chromosome aberration		

[4]

Recent research has shown that a gene on the X-chromosome may contribute to ovarian cancer predisposition. Having one copy of the mutant allele increases the risk of developing ovarian cancer, while having two copies of the mutant allele further increases the risk of developing ovarian cancer.

- (ii) Using your knowledge of cancer, explain why a female homozygous for the mutant allele does not necessarily develop cancer.

.....
.....
.....
.....[2]

- (iii) Using the symbols X^R to represent the normal allele and X^r to represent the mutant allele, draw a genetic diagram to show how a female homozygous for the normal allele will have all daughters being heterozygous for the mutant allele.

[3]

[Total: 26]

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2. The hedgehog, *Erinaceus europaeus*, is a small carnivorous mammal native to Northern Europe.

Hedgehogs were introduced onto a small group of islands off the west coast of Scotland in 1974. The hedgehog population has increased so that there are now over 5 000 breeding pairs. These hedgehogs have no natural predators on these islands and their diet consists mainly of bird's eggs.

Fig. 2.1 shows the distribution of *Erinaceus europaeus* population across the north and south island off the west coast of Scotland in the year 2000.

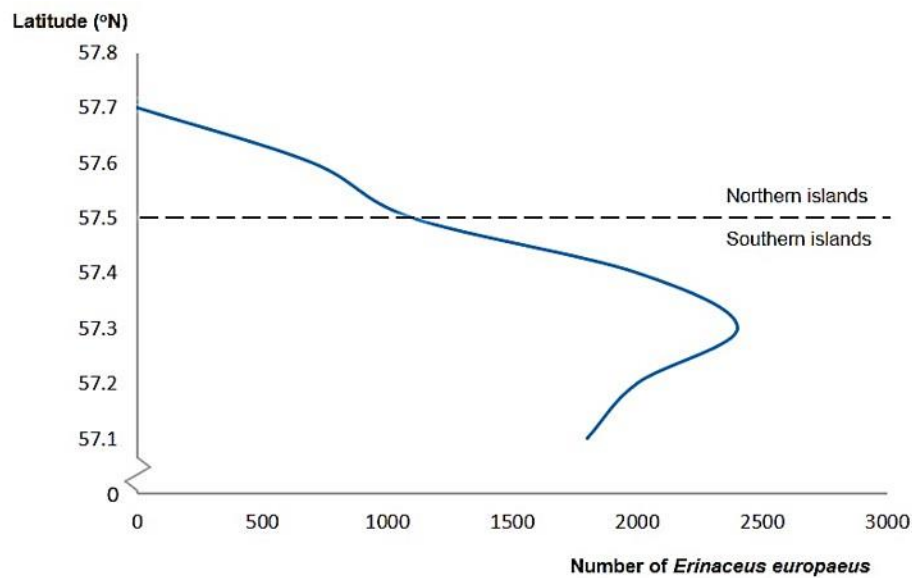


Fig. 2.1

Table 2.1 shows the changes in the population of the species of birds found on the same islands from 1983 to 2000.

Table 2.1

	breeding pairs in 1983	breeding pairs in 2000	% change in population
north island			
oystercatcher	928	1122	—
lapwing	1104	1364	+24
redshank	486	733	+51
south island			
oystercatcher	907	1403	—
lapwing	1869	1287	-31
redshank	1288	760	-41

- (a) (i) Calculate the % change in population of oystercatcher in the north and south island respectively. **Write your answers in Table. 2.1**, leave your answers to the **nearest percentage**.

[1]

- (ii) Using Fig. 2.1 and Table. 2.1, account for the difference in % change in population of **redshank**.

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.....[4]

- (iii) Suggest reasons for the increase in oystercatcher population on the south island.

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.....[2]

Variation plays an important role in the process of evolution by natural selection.

Gene mutation is one source of variation.

(b) Apart from gene mutation, outline one other source of variation and explain why variation is important in natural selection.

source of variation

why variation is important

.....

.....

.....

.....[3]

[Total: 10]

3. Maize, *Zea mays*, is a major cereal food crop.

The expected increase in carbon dioxide concentration in the atmosphere is expected to increase the rate of growth of many crop plants.

(a) (i) State the exact location in a plant cell where carbon dioxide is used.

.....[1]

Carbon dioxide is a gas.

(ii) Outline how plants use carbon dioxide for growth.

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.....[4]

Investigations were carried out into the effect of increased carbon dioxide concentration on the rate of photosynthesis in maize.

- Maize plants were grown in open-air trials, in the same field and were exposed to the same changes in the weather (with the exception of carbon dioxide concentrations).
- 50% of the plants were exposed to a normal carbon dioxide concentration.
- 50% of the plants were exposed to an increased carbon dioxide concentration.
- The rate of photosynthesis was measured as the net assimilation rate of carbon dioxide which indicates the overall amount of carbon dioxide taken in by the plant.
- Measurements were made at three-hourly intervals between 0700 hours and 1900 hours on three different days.

The results are shown in Fig. 3.1.

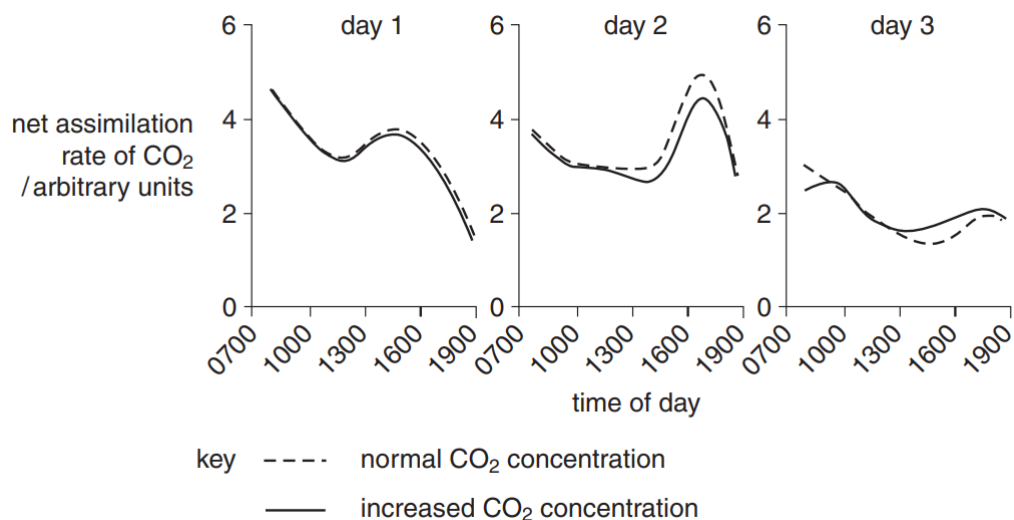


Fig. 3.1

(b) Account if the results shown in Fig. 3.1 supported the expected increase in rate of growth of crop plants in increased CO₂ concentration.

.....

.....

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.....[2]

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Plants under heat stress are exposed to high temperatures beyond a threshold level for a period of time that results in permanent damage to the plant's growth and development. While plants under dry stress lose excessive amount of water.

Both heat stress and dry stress can be attributed to climate change and one of the main drivers of climate change is the rise in atmospheric carbon dioxide levels.

Fig. 3.2 shows the predicted intensity of heat stress in Africa and dry stress in South America.

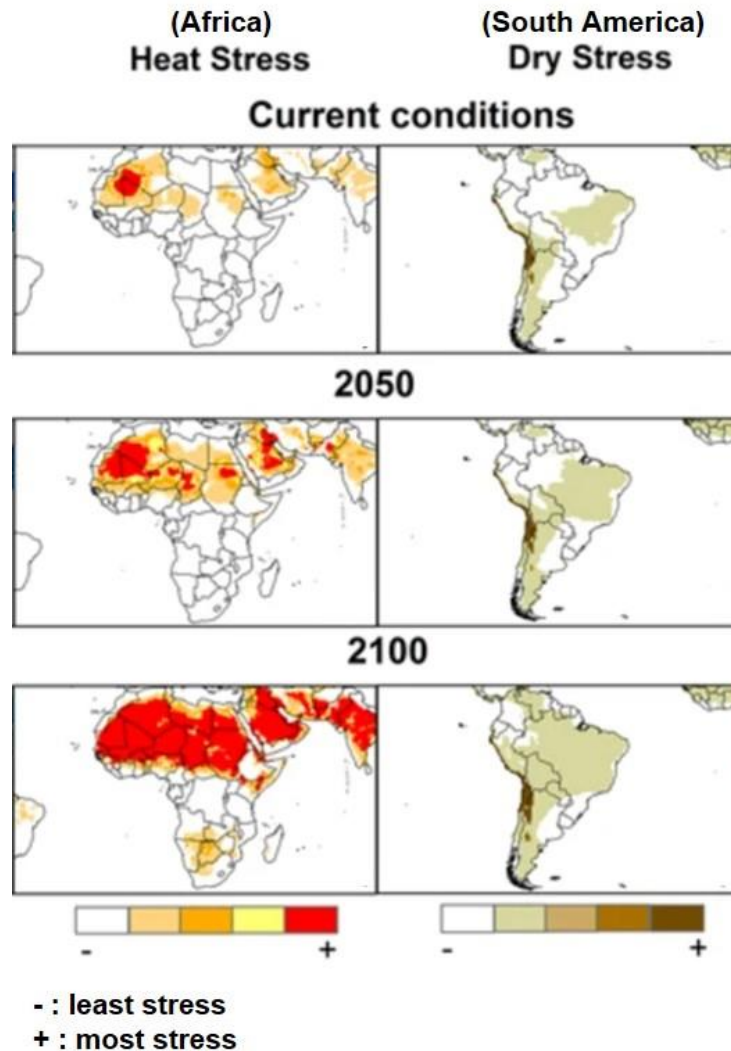


Fig. 3.2

The current distribution of maize crops are shown in Fig. 3.3. Computer modelling was used to predict the changes to distribution of maize crops in 2050 and 2100. Conditions are ranked from most optimal for growth to unsuitable for growth in the following order: Optimal > Suitable > Marginal > Unsuitable.

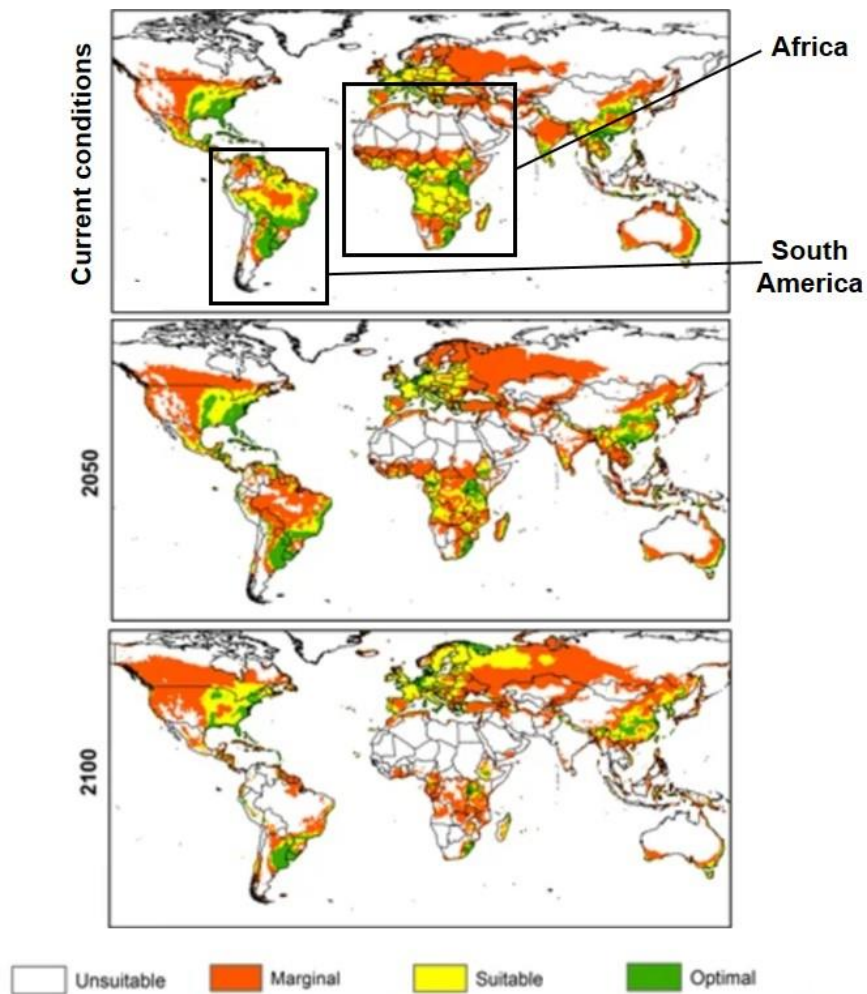


Fig. 3.3

(c) With reference to Fig. 3.2 and Fig. 3.3, account for the change in distribution of maize crops in South America.

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.....[2]

[Total: 9]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

4

- (a)** Describe the roles played by proteins in mitotic cell cycle.

[5]

- (b)** With reference to named examples, explain the importance of the structure of biopolymers to their functions.

[10]

[Total: 15]

5

- (a)** Describe the roles played by proteins in cellular respiration.

[5]

- (b)** Explain how mutations in stem cells can affect an individual and also subsequent generations.

[10]

[Total: 15]

8876/02/PU2 H1 End-of-Year Exams/2021